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Oefeningenreeks 6b nr. 17.2

$$\begin{aligned}s_n &= 1S_2(n) + 2S_1(n) \\&= \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n + 2\left(\frac{1}{2}n^2 + \frac{1}{2}n\right) \\&= \frac{2x^3 + 9n^2 + 7n}{6} \\s_n \in \mathbb{N} &\Leftrightarrow 2x^3 + 9n^2 + 7n = h \equiv 0(\text{mod } 6) \\n \equiv 0(\text{mod } 6) &\Rightarrow h \equiv 0(\text{mod } 6) \\n \equiv 1(\text{mod } 6) &\Rightarrow h \equiv 2 + 9 + 7 \equiv 18 \equiv 0(\text{mod } 6) \\n \equiv 2(\text{mod } 6) &\Rightarrow h \equiv 1 + 36 + 14 \equiv 66 \equiv 0(\text{mod } 6) \\n \equiv 3(\text{mod } 6) &\Rightarrow h \equiv 54 + 81 + 21 \equiv 156 \equiv 0(\text{mod } 6) \\n \equiv 4(\text{mod } 6) &\Rightarrow h \equiv 128 + 144 + 28 \equiv 300 \equiv 0(\text{mod } 6) \\n \equiv 5(\text{mod } 6) &\Rightarrow h \equiv 250 + 225 + 35 \equiv 510 \equiv 0(\text{mod } 6) \\&\text{dus } 2x^3 + 9n^2 + 7n \equiv 0(\text{mod } 6) \\&\text{dus } s_n \in \mathbb{N} \\&\text{QED}\end{aligned}$$

References